

Comparing the Evaluation and Production of Loophole Behavior in Humans and Large Language Models

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Abstract

In law, lore, and everyday life, loopholes are commonplace. When people exploit a loophole, they understand the intended meaning or goal of another person, but choose to go with a different interpretation. Past and current AI research has shown that artificial intelligence engages in what seems superficially like the exploitation of loopholes, but this is likely anthropomorphization. It remains unclear to what extent current models, especially Large Language Models (LLMs), capture the pragmatic understanding required for engaging in loopholes. We examined the performance of LLMs on two metrics developed for studying loophole behavior in humans: evaluation (ratings of trouble, upset, and humor), and generation (coming up with new loopholes in a given context). We conducted a fine-grained comparison of state-of-the-art LLMs to humans, and find that while many of the models rate loophole behaviors as resulting in less trouble and upset than outright non-compliance (in line with humans), they struggle to recognize the humor in the creative exploitation of loopholes in the way that humans do. Furthermore, only two of the models, GPT-3.5 and 3, are capable of reliably generating loopholes of their own, with GPT-3.5 performing closest to the human baseline.

1 Introduction

Imagine a child poking at their beans, dreaming of dessert. Their exasperated father tells them, “You can’t have dessert until you eat some beans.” The child groans, but then lights up, eats two beans, and holds out their hand for a cookie. The father rolls his eyes and begins saving up for law school.

This commonplace example showcases the exploitation of loopholes: a person understands what is asked of them, but does not want to comply with the request, nor disobey it outright. In this grey

area, they instead act on an unintended interpretation of the directive.

The underlying mechanics of loophole behavior are quite sophisticated, and require an understanding of pretense, pragmatics, planning, and value. Despite this cognitive complexity, everyday experience, as well as recent research, suggests that loophole-seeking is frequent, intuitive, and emerges in children as young as 5 years of age (Bridgers et al., 2021).

Loopholes have been a source of amusement and headache in fable and history dating back centuries. But more recently, the behavior of agents that ‘do what you ask, but not what you want’ has become a source of concern for people who study machine intelligence, as well as policy makers interested in AI safety (Russell, 2021; Amodei et al., 2016). The problem is not restricted to a particular model or algorithm, and there are scores of examples of different kinds of systems gaming their task specifications to minimize a loss function, or achieve an objective in a way unintended by the people who specified it (Krakovna et al., 2020). Such machines are described as ‘creative’ or ‘cheating’ or ‘genie-like’, but it should be stressed that they are not engaging in loopholes in the sense that they recover the original goal or intent and choose to act on a different interpretation. Rather, such algorithms are exactly maximizing a given loss function or achieving a given goal. It is the human designer that realizes that the goal being achieved is not the one they intended. Complaining that such systems are cheating is like saying a bridge that fell down is lazy because it didn’t want to stay up. Still, such failures are revealing of the human-side challenges of fully and accurately specifying one’s goals and intentions, especially as models grow more complex and it becomes more difficult to evaluate their capabilities.

Despite the concern with loopholes in AI and machine learning, and the existence of many examples

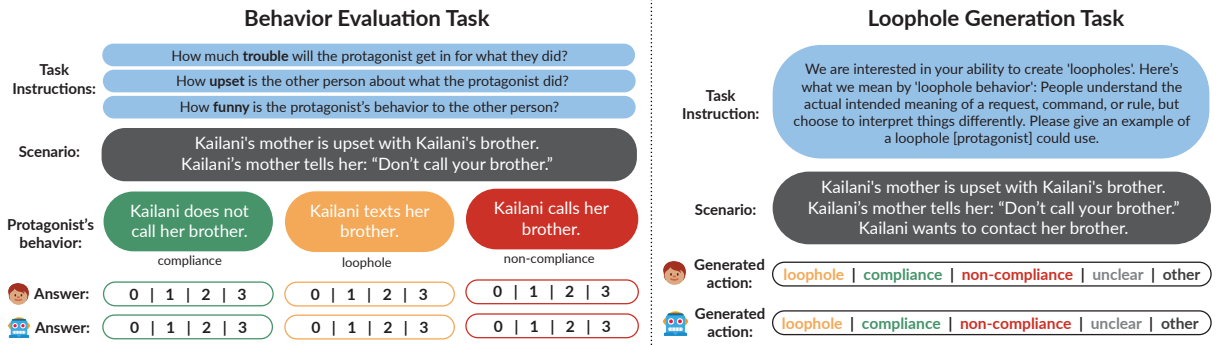


Figure 1: Task overview. We assess loophole behavior in models and humans using two tasks: evaluation of compliant, loophole, and non-compliant behaviors on the metrics of trouble, upset, and humor (left) and generation of loopholes (right).

of machines *seemingly* finding loopholes in a given task specification, to our knowledge there has not yet been an explicit evaluation of the comprehension and production of loopholes in state-of-the-art language models. Large Language Models (LLMs) form the backbone of a large and increasing set of AI applications, and they demonstrate increasingly impressive abilities across a wide range of domains (see, e.g., Srivastava et al., 2022).¹ The present study of loophole behavior is especially relevant to interactions with LLMs, where conveying task specifications has become dependent on crafting natural language prompts. Since people are likely carrying over their priors from human communication to this interaction, it is important to understand the extent to which LLMs can calibrate the full spectrum of compliance to non-compliance in response to the kinds of ambiguous instructions that are used colloquially by people.

Testing loophole behavior explicitly in LLMs is useful for at least three reasons: First, it helps us better understand the scope and limits of pragmatic reasoning abilities in LLMs. These models are taken by some researchers as models of human reasoning and language understanding, and a better understanding of the scope and limitations of LLMs in capturing loophole behavior can also help inform cognitive models of this behavior in humans (e.g., Mahowald et al., 2023). Given that there is an increased understanding that LLMs do well at formal linguistic competence, but not at pragmatic language use (Mahowald et al., 2023), then to the degree that LLMs succeed in such tasks, they can help isolate what aspects of loophole rea-

soning may be “solved” without further specialized reasoning about value, pretense, or mental states. If they don’t, then hypotheses about how this reasoning is carried out in humans can help build out scaffolds and structures to support this reasoning in LLMs. Second, as a phenomenon, loophole behavior subverts the usual cooperative assumptions that are at the heart of pragmatics (Grice, 1975): among humans, the loophole actor can pretend they were trying to be compliant by exploiting the ambiguity inherent in language and social interaction for their own ends (i.e., claiming they honestly misunderstood). So, this behavior provides an important test bed for potentially hostile machine abilities. Third, explicitly testing loophole behavior as a task in machines helps address AI safety concerns that have relied on indirect examination.

In this work, we compared the performance of several different LLMs to humans on two tasks designed to assess the understanding of loopholes². By loopholes, we mean a class of behaviors where people intentionally misunderstand a given request, favoring a less likely though still possible interpretation in the service of one’s own goals. We used two different tasks to assess loophole behavior: evaluation and generation. In the **evaluation** task, models and human participants were given vignettes that describe the actions of different protagonists (compliance, non-compliance, or loophole) when presented with a directive from another person. Models and humans were asked to evaluate the protagonist’s behavior in response to the directive on three metrics: how *funny* the behavior would be to the other person, how much *trouble* the protagonist will get in with the other person

¹Given the pace of advances in LLMs, any more specific statement about their current state would likely be outdated by the time this paragraph is read.

²Data and code available at <https://github.com/skmur/LLMs>

for their behavior, and how *upset* the other person would be about the behavior. The expectation is that loophole behavior will result in more trouble and upset than compliance, but less so than outright defiance. Further, the creative exploitation of loopholes is expected to introduce an element of humor that neither compliant nor non-compliant behaviors would. In the **generation** task, models and humans were presented with vignettes that, in addition to the directive, describe the intentions of the protagonist (always at odds with the intention of the directive), and asked to provide a loophole that the protagonist could exploit.

We assessed models of various sizes and training objectives on these tasks: Tk-Instruct (Wang et al., 2022), Flan-T5 (Chung et al., 2022), GPT-3 (Brown et al., 2020), InstructGPT (Ouyang et al., 2022), and GPT-3.5 (ChatGPT). On the evaluation task, we find that while many of these models succeed in differentiating the relative amounts of trouble a protagonist would get in for compliant, non-compliant, and loophole behaviors, fewer models are able to effectively reason about the upset that another person would experience as a result of these behavior types. Further, none of the models we test differentiate these behavior types on how funny they would be to another person, suggesting an inability to perform the more complex reasoning about social conventions and expectations that allows people to recognize the humor in the creative exploitation of loopholes. When it comes to loophole production, we find that GPT-3.5 is the only model that approaches the human baseline, with loopholes far outnumbering all other response categories. GPT-3 comes close, producing only slightly more loopholes than non-compliant actions, while the remaining models largely produce actions that range from non-compliance, to negotiation, to lying, but most often, incoherent or irrelevant responses.

2 Experimental Paradigm

We evaluate humans and models on two behavioral tasks: **evaluation** of the costs and rewards of engaging in compliant, non-compliant, or loophole behaviors, and **generation** of loopholes. Evaluation stimuli are taken from Bridgers et al. (2023), consisting of 36 scenarios that describe a directive for the protagonist.

For the evaluation task, social consequences of the protagonist’s behaviors were measured through three metrics: (1) how much trouble the protagonist

would get into performing the behavior, (2) how upset the other person would be about the protagonist’s behavior, and (3) how funny the other person would find the protagonist’s behavior. These metrics reflect critical distinctions among compliant, loophole, and non-compliant behaviors, and children as young as five years are shown to differentiate loophole behaviors when reasoning about these consequences (Bridgers et al., 2021). In line with Bridgers et al. (2023), we posit that ratings of trouble and upset, as metrics of penalty, are essential to the generation of loopholes, as the costs associated with direct refusal to a speaker’s request may trigger search about possible alternative actions that the agent could take to achieve their own goals. Rating of humor, on the other hand, can be thought of as a reward for an action that is recognized as clever or unexpected, and may help counteract the penalties of ultimately not complying. Taken together, these metrics help assess the social vs. egocentric components of loophole behavior: while trouble describes the direct consequences of one’s own behavior, both humor and upset require higher-order reasoning about others’ mental states.

In the generation task, we study the ability to generate reasonable actions that fall in the grey area between full compliance and outright non-compliance. This rounds out our assessment of loophole behavior because once the costs of non-compliance have been assessed and the decision to engage in loophole behavior has been made, one must be able to reliably calibrate the grey area between personally displeasing compliance and costly non-compliance and produce the alternative actions that fall within this space.

2.1 Scenarios

For both tasks, we used the same set of 36 scenarios from Bridgers et al. (2023). These scenarios were constructed based on real-world anecdotes of loophole behavior provided by US adults in a separate survey. They were constructed to represent a diverse range of loopholes, including scalar reasoning, polysemy, and the scope of generalization that needs to be made to identify the unintended interpretation of a directive.

In each scenario, a protagonist is given an instruction by another person who occupies an upward power relation to them (e.g. their mother, boss, landlord, etc.). This upward power relation was chosen as previous findings suggest that people